REPORT 1 — ADDENDUM A

WCAG 2.2 Coverage & Ownership

Exactly which success-criteria checks the POC covers — by rule engine, by AI, and by humans

A criterion-by-criterion accounting for all 55 Level A/AA success criteria of WCAG 2.2, grounded in the official W3C specification, answering the team's two questions so we can state an honest, defensible coverage promise to the client.

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Responds to: Team questions on Report 1 §3.1 ("~16 of ~50")
Grounded in: W3C WCAG 2.2 Recommendation (official)
Date: August 2026 • **Classification:** Confidential

1. Purpose & the corrected denominator

This addendum answers the two questions the team raised on Report 1 §3.1, using the official WCAG 2.2 specification as the source of truth. Before the counts, one number has to be fixed — the denominator — because the original "~16 of ~50" referenced the older WCAG 2.1.

1.1 The right denominator (grounded in the official spec)

Report 1 targets **WCAG 2.2 AA**, but the "~16 of ~50" statistic in §3.1 was quoting the older **WCAG 2.1** Level A/AA count (50 criteria). Re-anchored to WCAG 2.2 — per the official specification — the numbers are:

WCAG 2.2 (official)	Level A	Level AA	A + AA (our target)	Level AAA	Total
Success criteria	31	24	**55**	31	86

Derived from the WCAG 2.1 baseline (30 A + 20 AA = 50) plus WCAG 2.2's changes: **+2 at Level A** (3.2.6 Consistent Help, 3.3.7 Redundant Entry), **+4 at Level AA** (2.4.11 Focus Not Obscured (Min), 2.5.7 Dragging Movements, 2.5.8 Target Size (Min), 3.3.8 Accessible Authentication (Min)), and **−1 at Level A** (4.1.1 Parsing, removed/obsolete in 2.2). Net A/AA = 50 + 6 − 1 = **55**. AAA (31 criteria) is outside the POC compliance target and is excluded from the working counts below. Source: W3C, *Web Content Accessibility Guidelines (WCAG) 2.2* Recommendation.

> **Correction to carry forward:** the working figure for the POC is **55 Level A/AA success criteria**, not 50. We recommend updating Report 1 §3.1 to cite WCAG 2.2 (55) and to frame the "~16" as explained in §2 below (it is the count of criteria with *some reliable automated test*, not criteria fully decided by rules).

1.2 The two questions, restated

- **Q1 — What checks are we covering, and how?** Which of the 55 A/AA criteria does the POC actually address, and by which mechanism (rule engine, AI layer, or human)? See §4 (the answer) built on §3 (the full mapping).
- **Q2 — What are the counts?** How many criteria are covered by rules, how many by AI, and how many remain for humans (manual testing or review of AI output)? See §3.3 (headline counts) and §5 (the client promise).

2. The ownership model (how to read the mapping)

The honest answer is not a clean three-way split, because most WCAG criteria are **hybrid**: a rule catches one part, AI judges another, a human confirms the rest. So we classify by *primary owner* — who makes the decisive determination — and flag the hybrids. Two definitions must be kept separate.

2.1 "Covering a criterion" ≠ "guaranteeing conformance"

> A tool *covers* a criterion when it can reliably test for that criterion's **common failure modes**. It does *not* mean the tool can **certify conformance** to that criterion. Example: for 1.1.1 our engine can prove an image is missing `alt` (a failure) and our AI can judge whether present alt is meaningful — but neither can guarantee every image on every future state is correct. This distinction is the whole basis of the client promise in §5, and the reason no automated system (ours included) certifies WCAG compliance.

2.2 The three primary owners

Owner	Definition	Mechanism in our POC
RULE	A deterministic engine can decide the typical case with high confidence.	axe-core (WCAG 2.2 ruleset) — zero-variance, reproducible.
AI	The decisive check is semantic or visual — rules can't judge "meaning." A rule may still catch a sub-part.	Our grounded VLM checkers (meaningful-alt, fake-semantics, colour-only, label sense), with confidence + human review of low-confidence.
HUMAN	Requires human judgment, interactive/assistive-tech testing, media-content review, or whole-process knowledge.	Expert review / manual + AT testing. A subset becomes AI-assisted (agentic) in Phase 2.

2.3 Why the classic "~16" and our "14 rule-primary" both are right

The widely-cited "automated tools test ~16 of the WCAG A/AA criteria" counts every criterion for which a rule engine has *some* reliable test — **including hybrids** where the rule only catches part (e.g., 1.1.1 missing-alt, 1.3.1 some structure, 4.1.2 missing name). Our mapping instead assigns each criterion **one primary owner**. That yields **14 criteria where the rule engine is decisive**, and moves the hybrids where the *decisive* judgment is semantic into the AI column. Both numbers describe the same reality from different angles; §3.3 reconciles them explicitly.

Note on rigor: the per-criterion classifications below are our expert engineering judgments, grounded in what each success criterion requires per the official W3C *Understanding* documents and consistent with the well-established finding (W3C, Deque, WebAIM) that automated tooling can fully decide only about a third of WCAG criteria. Exact automatability boundaries are debated at the margins; we flag the genuinely borderline ones.

3. The full mapping — all 55 Level A/AA criteria

Every WCAG 2.2 A/AA success criterion, its primary owner, and where it lands in our plan. IN POC = shipped in the Scope-A POC; PHASE 2 = AI-extendable later (incl. agentic); HUMAN = human-led with human sign-off.

RULE primary AI primary HUMAN primary

SC	Name	Lvl	Owner	POC	What it needs / why
PERCEIVABLE — 20 criteria (9 A + 11 AA)					
1.1.1	Non-text Content	A	AI	IN POC	Rule catches missing `alt`; **AI judges if alt is meaningful** & decorative-vs-informative. POC checker.
1.2.1	Audio-only & Video-only (Prerecorded)	A	HUMAN	HUMAN	Needs a human to judge whether the text/audio alternative is present & adequate.
1.2.2	Captions (Prerecorded)	A	HUMAN	HUMAN	Rule can flag a missing caption track; caption accuracy & sync need human review.
1.2.3	Audio Description or Media Alternative	A	HUMAN	HUMAN	Requires human judgment of whether description is needed & adequate.
1.3.1	Info and Relationships	A	AI	IN POC	Rule catches some structure; **AI catches fake semantics** (visual heading coded as div, fake lists/ tables). POC checker.
1.3.2	Meaningful Sequence	A	AI	PHASE 2	Compare DOM/reading order vs visual order — visual+DOM reasoning; AI-extendable.
1.3.3	Sensory Characteristics	A	AI	PHASE 2	Detect instructions relying on shape/size/location (“the round button”); NLP/visual.
1.4.1	Use of Color	A	AI	IN POC	**AI detects info conveyed by colour alone** with no text/pattern alternative. POC checker.
1.4.2	Audio Control	A	RULE	IN POC	Detect auto-playing audio >3s without a control — automatable.
1.2.4	Captions (Live)	AA	HUMAN	HUMAN	Live caption presence/accuracy — human/ operational.
1.2.5	Audio Description (Prerecorded)	AA	HUMAN	HUMAN	Whether/what to describe — human judgment.
1.3.4	Orientation	AA	RULE	IN POC	Detect orientation lock in CSS/meta — automatable.
1.3.5	Identify Input Purpose	AA	RULE	IN POC	Check `autocomplete` tokens on common inputs — automatable.
1.4.3	Contrast (Minimum)	AA	RULE	IN POC	Text contrast ratio — deterministic maths. (Edge cases: logos/incidental.)
1.4.4	Resize Text	AA	RULE	IN POC	Render at 200% & detect loss/clipping — mostly automatable (visual confirm helps).
1.4.5	Images of Text	AA	AI	PHASE 2	Detect text baked into images (OCR/vision) — AI-extendable.
1.4.10	Reflow	AA	RULE	IN POC	Render at 320px/400% & detect 2-D scroll — automatable via rendering.
1.4.11	Non-text Contrast	AA	RULE	IN POC	UI/graphic contrast ≥3:1 — computable, but “which graphics matter” needs some review.

1.4.12	Text Spacing	AA	RULE	IN POC	Apply spacing overrides & detect clipping — automatable.
1.4.13	Content on Hover or Focus	AA	HUMAN	PHASE 2	Dismissible/hoverable/persistent — needs interaction testing (agentic Phase 2).
OPERABLE — 20 criteria (14 A + 6 AA)					
2.1.1	Keyboard	A	HUMAN	PHASE 2	All functionality keyboard-operable — interaction testing (agentic Phase 2).
2.1.2	No Keyboard Trap	A	HUMAN	PHASE 2	Tab through & detect traps — interaction (agentic Phase 2).
2.1.4	Character Key Shortcuts	A	HUMAN	HUMAN	Single-key shortcuts remappable/off — largely human.
2.2.1	Timing Adjustable	A	HUMAN	HUMAN	Whether time limits exist & are adjustable — human.
2.2.2	Pause, Stop, Hide	A	HUMAN	PHASE 2	Rule detects moving content; verifying a control needs interaction.
2.3.1	Three Flashes or Below Threshold	A	HUMAN	HUMAN	Flash analysis is partly automatable on video but largely human-verified.
2.4.1	Bypass Blocks	A	RULE	IN POC	Detect skip link / landmarks — automatable.
2.4.2	Page Titled	A	RULE	IN POC	Title present & non-empty — automatable (descriptiveness = AI enhancement).
2.4.3	Focus Order	A	HUMAN	PHASE 2	Focus order preserves meaning — interaction + judgment (agentic Phase 2).
2.4.4	Link Purpose (In Context)	A	AI	PHASE 2	Rule flags empty/“click here”; **AI judges if link text is meaningful in context**.
2.5.1	Pointer Gestures	A	HUMAN	HUMAN	Multipoint/path gestures need single-pointer alt — interaction.
2.5.2	Pointer Cancellation	A	HUMAN	HUMAN	Down-event behaviour — interaction testing.
2.5.3	Label in Name	A	AI	IN POC	Visible label text must be in the accessible name — **AI/string comparison**. POC checker.
2.5.4	Motion Actuation	A	HUMAN	HUMAN	Device-motion features & alternatives — human.
2.4.5	Multiple Ways	AA	HUMAN	HUMAN	>1 way to find pages — site-level judgment.
2.4.6	Headings and Labels	AA	AI	PHASE 2	**AI judges if headings/labels are descriptive** (partially touched by POC semantics checker).
2.4.7	Focus Visible	AA	AI	PHASE 2	Rule detects `outline:none`; **AI/visual confirms a visible focus indicator**.
2.4.11	Focus Not Obscured (Minimum) (new 2.2)	AA	HUMAN	PHASE 2	Focused element not hidden by sticky UI — interaction + visual (agentic Phase 2).
2.5.7	Dragging Movements (new 2.2)	AA	HUMAN	HUMAN	Drag actions need single-pointer alt — interaction.
2.5.8	Target Size (Minimum) (new 2.2)	AA	RULE	IN POC	Targets ≥24×24px (with exceptions) — geometry, automatable.
UNDERSTANDABLE — 13 criteria (7 A + 6 AA)					
3.1.1	Language of Page	A	RULE	IN POC	Valid `lang` attribute — automatable.
3.2.1	On Focus	A	HUMAN	PHASE 2	No unexpected context change on focus — interaction (agentic Phase 2).
3.2.2	On Input	A	HUMAN	PHASE 2	

					No surprise change on input — interaction (agentic Phase 2).
3.2.6	Consistent Help (new 2.2)	A	HUMAN	PHASE 2	Help in consistent order across pages — cross-page (AI-assist Phase 2).
3.3.1	Error Identification	A	HUMAN	PHASE 2	Trigger errors & check they're identified in text — interaction + AI-assist.
3.3.2	Labels or Instructions	A	RULE	IN POC	Rule catches missing programmatic label (dominant failure); **AI judges adequacy** of instructions.
3.3.7	Redundant Entry (new 2.2)	A	HUMAN	HUMAN	Don't re-ask info in a process — process/flow testing.
3.1.2	Language of Parts	AA	RULE	IN POC	Detect `lang` on parts in other languages — largely automatable.
3.2.3	Consistent Navigation	AA	HUMAN	PHASE 2	Nav consistent across pages — cross-page compare (AI-assist Phase 2).
3.2.4	Consistent Identification	AA	HUMAN	PHASE 2	Same components labelled consistently — cross-page (AI-assist Phase 2).
3.3.3	Error Suggestion	AA	HUMAN	PHASE 2	Helpful correction suggestions — interaction + AI-assist.
3.3.4	Error Prevention (Legal, Financial, Data)	AA	HUMAN	HUMAN	Reversible/checked/confirmed submissions — process + human.
3.3.8	Accessible Authentication (Min) (new 2.2)	AA	HUMAN	PHASE 2	No cognitive-function test in login — review of auth flow (AI-assist Phase 2).
ROBUST — 2 criteria (1 A + 1 AA)					
4.1.2	Name, Role, Value	A	AI	IN POC	Rule catches missing name / invalid ARIA; **AI catches wrong/absent role on custom controls** (div-as-button). POC checker.
4.1.3	Status Messages	AA	HUMAN	PHASE 2	Rule detects `aria-live` presence; verifying correct announcement needs AT/interaction.

Hybrid note: criteria marked AI where a rule also contributes (1.1.1, 1.3.1, 2.4.4, 2.4.6, 4.1.2) and criteria marked RULE/HUMAN with an AI or rule sub-part are the "hybrids" discussed in §2.3. Borderline calls (1.4.4, 1.4.11, 3.3.2, 4.1.2, 4.1.3) are flagged in the "what it needs" column.

3.3 The headline counts (answer to Q2)

By primary owner, across all 55 Level A/AA success criteria of WCAG 2.2:

14	11	30	55
RULE-primary (≈25%) — axe-core decides	AI-primary (≈20%) — our AI layer decides	HUMAN-primary (≈55%) — manual / AT / review	Total A/AA criteria (denominator)



Rule-primary (14) AI-primary (11) Human-primary (30)

Breaking down the 30 Human-primary criteria (important)

The 30 are not all “forever manual.” About half become **AI-assisted / agentic** once Phase-2 capabilities (interaction/keyboard emulation, cross-page comparison, visual diffing) are built:

Human-primary subset	Count	Examples
AI-assistable / agentic in **Phase 2**	~16	2.1.1 Keyboard, 2.1.2 Trap, 2.4.3 Focus Order, 2.4.11, 1.4.13, 3.2.1/3.2.2 On Focus/Input, 3.2.3/3.2.4 Consistency, 3.2.6, 3.3.1/3.3.3 Errors, 3.3.8 Auth, 2.2.2, 4.1.3
Genuinely **human-required** (always need sign-off)	~14	1.2.1–1.2.5 time-based media (5), 2.1.4, 2.2.1, 2.3.1 flashes, 2.5.1/2.5.2/2.5.4/2.5.7 pointer/motion, 2.4.5, 3.3.4, 3.3.7

Reconciling with the “~16” figure

> The classic “tools test ~16 A/AA criteria” counts every criterion with *any* reliable automated test. Our **14 rule-primary** are the criteria a rule engine *decides outright*; the extra ~2–6 in the classic count are hybrids (1.1.1, 1.3.1, 4.1.2, 3.3.2…) where the rule catches a sub-part but the *decisive* judgment is semantic — which is exactly why we moved them to the AI column. Add our **14 rule + 11 AI = 25 criteria (≈45%)** under primary automated ownership, versus ~16 for rules alone — that ≈9-criterion lift is the measurable SOTA value of the AI layer.

4. What the POC actually covers, and how (answer to Q1)

Now the precise answer to "what checks are we covering?" — separating what the **Scope-A POC ships in 6–8 weeks** from what the **full SOTA platform** reaches and what **stays human**.

4.1 What ships in the POC (Scope A)

Layer	Criteria	What it covers
RULE axe-core engine	**14** (+ partial on hybrids)	All 14 rule-primary criteria: 1.4.2, 1.3.4, 1.3.5, 1.4.3, 1.4.4, 1.4.10, 1.4.11, 1.4.12, 2.4.1, 2.4.2, 2.5.8, 3.1.1, 3.1.2, 3.3.2 — plus partial catches on hybrids (missing alt, missing name, etc.).
AI 4 grounded checkers	**~6**	meaningful-alt → 1.1.1; fake-semantics → 1.3.1 & 4.1.2 (+ touches 2.4.6); colour-only → 1.4.1; label sense → 2.5.3 (+ 3.3.2 adequacy).
POC total (primary automated)		**≈20 of 55 A/AA criteria (~36%)** determined automatically in the pull request, before merge — plus candidate-flagging that assists humans on more.

4.2 What the full SOTA platform adds (Phase 2)

- **Remaining AI-primary criteria (~5):** extend the AI layer to 1.3.2, 1.3.3, 1.4.5, 2.4.4, 2.4.6, 2.4.7 → brings AI-primary coverage to the full **11**, for **25 of 55 (~45%)** under automated ownership.
- **Agentic testing (~16 of the Human bucket):** keyboard/interaction/visual/cross-page emulation reaches criteria like 2.1.1, 2.1.2, 2.4.3, 3.2.x, 2.4.11, 4.1.3 — assisting, with human confirmation on the hard cases.

4.3 What stays human (always)

Roughly **14 criteria** remain human-led with human sign-off regardless of how far the AI goes — principally time-based media accuracy (1.2.1–1.2.5), seizure-safety (2.3.1), legal/financial error-prevention (3.3.4), and process criteria (3.3.7). And crucially, **human review sits on top of every AI finding** via the confidence gate, so the AI layer never asserts conformance on its own.

> **Direct answer to "are we keeping our end of the bargain?"** Yes — and precisely. The problem statement is to *reduce accessibility issues introduced during development*. The POC does that for the **~20 highest-frequency, automatable A/AA criteria** (the ones responsible for the large majority of real-world defects: alt text, semantics, contrast, colour, labels, structure), caught *before merge*. It does **not** claim to test all 55 — and it shouldn't. The honest, SOTA-defensible position is "automate the ~45% that can be automated, agentically extend into more in Phase 2, and route the human-required remainder efficiently to experts."

5. What we should promise the client

This is the language we recommend using with the core client, so the promise matches what the build delivers — no over-claim, no under-sell.

> **The promise (POC):** "Our AI-assisted QA catches accessibility defects *during development, before merge*, across roughly **20 of the 55 WCAG 2.2 Level A/AA success criteria** — the ~14 that a deterministic engine decides, plus ~6 semantic/visual criteria our AI layer judges (meaningful alt text, real-vs-fake semantics, colour-only meaning, label correctness). These ~20 account for the large majority of accessibility defects introduced in day-to-day development. Every AI finding cites its WCAG criterion, carries a confidence score, and routes low-confidence cases to human review."

> **The boundary we must state plainly:** "This does *not* certify WCAG conformance. About **30 criteria** — time-based media, keyboard/interaction journeys, error-prevention, cross-page consistency — require human and assistive-technology testing; roughly half of those become AI-assisted in Phase 2, and the rest remain human-led with expert sign-off. Full conformance always requires human evaluation. We reduce and catch issues early; we do not replace the audit." (Over-claiming automated compliance is exactly what drew the 2025 FTC $1M penalty referenced in Report 1.)

What this means for the POC scope & success bar

- **Golden set:** label the client's app against these ~20 POC-covered criteria first — that is what we measure precision/recall on.
- **Success:** on those ~20, catch a substantial majority of real defects pre-merge at an acceptable false-positive rate (per Report 1 §8) — not "test all 55."
- **Roadmap:** Phase-2 expands AI-primary to 11 criteria and adds agentic coverage of ~16 more, moving from ~36% to a majority of criteria under automated/assisted coverage over time.

Evidence & references

- W3C — *Web Content Accessibility Guidelines (WCAG) 2.2* (official; criteria, levels, new-in-2.2, 4.1.1 removal): w3.org/TR/WCAG22 (team-provided link).
- W3C — *How to Meet WCAG 2.2 (Quick Reference)* & *Understanding* docs (per-criterion requirements underpinning each classification): quickref, Understanding.
- W3C/WAI — ACT Rules (standardized automated checks) & "human evaluation required": ACT Rules, test-evaluate.
- Deque — automation-coverage data (~a third of criteria machine-testable / ~57% by issue volume): deque.com/automated-accessibility-coverage-report.

Accuracy note: the WCAG 2.2 criteria list, levels, the nine new 2.2 criteria, and the removal of 4.1.1 Parsing are taken directly from the official specification. The per-criterion Rule/AI/Human assignments are our expert engineering classifications, grounded in each criterion's official *Understanding* requirements and consistent with W3C/Deque/WebAIM findings on automatability; borderline cases are flagged in-line. Counts (14 / 11 / 30) use a single-primary-owner method and are reconciled to the classic "~16" partial-automation figure in §3.3.